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USER AUTHORIZATION FOR FILE LEVEL RESTORATION FROM IMAGE LEVEL BACKUPS

Abstract

Embodiments provide systems, methods, and computer program products for enabling user authorization to perform a file level recovery from an image level backup of a virtual machine without the need for access control by an administrator. Specifically, embodiments enable an access control mechanism for controlling access to stored image level backups of a virtual machine. In an embodiment, the virtual machine includes a backup application user interface that can be used to send a restoration request to a backup server. The restoration request can include a machine identifier and a user identifier of the user logged onto the virtual machine. The backup server includes a backup application that determines whether or not the machine identifier contained in the restoration request can be matched to a machine identifier of a virtual machine present in one of the virtual machine backups stored on the backup server.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/387,727, filed on Nov. 7, 2023, which is a continuation of U.S. application Ser. No. 17/329,732 filed on May 25, 2021, which is now U.S. Pat. No. 11,860,738, which is a continuation of U.S. application Ser. No. 16/552,720 filed on Aug. 27, 2019, now U.S. Pat. No. 11,016,856, which is a continuation of U.S. application Ser. No. 14/831,576 filed on Aug. 20, 2015, now U.S. Pat. No. 10,394,657, issued Aug. 27, 2019, which claims the benefit of U.S. Provisional Application No. 62/040,295 filed on Aug. 21, 2014, which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to image level data protection, and more particularly to providing user authorization during self-service file-level restoration from image level backups and replicas.

BACKGROUND

Background Art

[0003] To prevent information loss, organizations generally implement a recovery plan that enables periodic backup of physical and virtual machines, and recovery of entire machines, individual data files, folders and application items.

Description

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

[0004] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the disclosure and to enable a person skilled in the pertinent art to make and use the disclosure.

[0005] FIG. 1 illustrates an example environment in which embodiments can be practiced or implemented.

[0006] FIG. 2 illustrates an example system according to an embodiment.

[0007] FIG. 3 is an example process according to an embodiment.

[0008] FIG. 4 illustrates another example system according to an embodiment.

[0009] FIG. **5** is another example process according to an embodiment.

[0010] FIG. **6** illustrates another example system according to an embodiment.

[0011] FIG. **7** is another example process according to an embodiment.

[0012] FIG. **8** illustrates another example system according to an embodiment.

[0013] FIG. **9** is another example process according to an embodiment.

[0014] FIG. **10** is an example computer system in which embodiments can be practiced or implemented.

[0015] The present disclosure will be described with reference to the accompanying drawings. Generally, the drawing in which an element first appears is typically indicated by the leftmost digit(s) in the corresponding reference number.

DETAILED DESCRIPTION OF EMBODIMENTS

[0016] According to some approaches, an image-level backup may be used to backup a physical or virtual machine (VM). This description will use a virtual machine as an non-limiting example. However, as would be understood by a person of skill in the art, embodiments are not limited to the use with virtual machines and may also be used with physical machines, for example. In case of VM, this may be accomplished by using a hypervisor snapshot (VM snapshot) of the VM. The VM snapshot may be used to create a complete copy of a VM image for backup purposes.

[0017] Existing backup solutions, such as Veeam Backup & Replication, enable restoring both the whole VM image, and individual files from image level backups without restoring the whole VM image.

[0018] FIG. **1** illustrates an example environment **100** in which embodiments can be practiced or implemented. Example environment **100** is provided for the purpose of illustration only and is not limiting of embodiments. As shown in FIG. **1**, example environment **100** includes a user system **102**, a host system **104**, and a backup server **106**. Host system **104** is connected to user system **102** via an interface **108** and to backup server **106** via an interface **116**.

[0019] User system **102** can be any computing device, such as a personal computer, smartphone, or tablet, to name a few examples. Host system **104** can be a server that enables various services and applications. In one embodiment, host system **104** includes a hypervisor **110**, which enables a plurality of virtual machines **112.1**, **112.2**, . . . , **112.N**. In an embodiment, hypervisor **110** includes a virtual machine monitor configured to launch and run virtual machines. Hypervisor **110** can be implemented in software, hardware, or firmware. For example, hypervisor **110** can be a VMware® vSphere hypervisor or a Microsoft® Hyper-V hypervisor.

[0020] Backup server **106** is a physical or virtual machine that performs the role of a configuration and control center, performing various types of administrative activities (such as coordinating backup, replication tasks, controlling backup scheduling and resource allocating).

[0021] In addition, backup server **106** can be used to store image-level backups. For example, as shown in FIG. **1**, backup server **106** can store image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N** respectively. Alternatively, image level backups can be stored on another server, network attached storage device (NAS), or storage area network (SAN).

[0022] A user of user system **102** can have access privileges to one or more of virtual machines **112.1**, **112.2**, . . . , **112.N**. For example, if the user of user system **102** has access privileges to virtual machine **112.1**, the user of user system **102** can log onto virtual machine **112.1** by providing its user credentials (login, password) to virtual machine **112.1**.

[0023] At times, the user of user system **102** may need to restore data from an image level backup of a virtual machine that it has access to. This is typically done by restoring an image (or a portion thereof) of the virtual machine from a backup file. In some systems, data restoration privileges are either subject to no restrictions, or limited to backup administrators only. The former approach creates security and privacy concerns since individual data, which may contain personal information, can be accessed/restored from an image level backup of the virtual machine by more than one user. The latter approach can create a large work load for administrators. For example,

administrators typically respond to a user restoration request by performing the restoration themselves. This can be time consuming, and also inefficient as it requires the availability of the administrator when the user has a need for restoring, and that the administrator is able to locate the required file within the user folders. Other approaches include creating a restoration team dedicated for handling restoration requests. However, this approach can be expensive and can create a privacy issue since the personnel of the restoration team will be able to access the user's contents of the image level backup of the virtual machine.

[0024] Embodiments, as further described below, include but are not limited to systems, methods, and computer program products for enabling user authorization during file level restore from an image level backup without the need for access control by a backup administrator. Specifically, example embodiments enable an access control mechanism for controlling access to stored virtual machine images by users in the system. In an embodiment, the virtual machine includes a backup application user interface that can be used to send a restoration request to a backup server. The restoration request can include machine identifier and a user identifier of the user logged onto the virtual machine. The machine identifier can be a DNS name, IP address assigned to virtual machine, hypervisor-level VM identifier, and similar. In another embodiment, machine identifier can be automatically detected by matching computer account (such as Active Directory computer account) to a VM. The user identifier can be the login information of the user logged onto the virtual machine, user token, or similar. The backup application can access and restore data from virtual machine backups, and can accept or deny the restoration request based on certain one or more criteria, for example whether or not machine identifier contained in restoration request can be matched to the machine identifier of virtual machine present in one of the virtual machine backups stored on backup server, or the user belongs to the certain access control group in the OS of virtual machine. In an embodiment, the access control group is a local administrators group.

[0025] FIG. 2 illustrates an example system **200** according to an embodiment. Example system **200** is provided for the purpose of illustration only, and is not limiting of embodiments. As shown in FIG. 2, example system **200** includes a user system **102**, a host system **104**, and a backup server **106**. Host system **104** includes a hypervisor **110**, which enables one or more virtual machines, such as virtual machine **112.1**. Host system **104** is connected to user system **102** via an interface **108** and to backup server **106** via an interface **116**.

[0026] In the example embodiments, backup server **106** also stores image level backups for one or more virtual machines on locally attached storage. For example, backup server **106** stores virtual machine image level backup VM_1 Backup **114.1** of virtual machine **112.1**. In addition, backup server **106** hosts a backup application **204**. Backup application **204** can be used by users to backup virtual machines, such as virtual machine **112.1**, to backup server **106** or to restore virtual machine images or portions thereof from virtual machine image level backup VM_1 Backup **114.1** stored on backup server **106**.

[0027] In an embodiment, during backup or replication activity, backup application **204** saves to a database user accounts belonging to a certain access control group in the Operating System (OS) of Virtual Machines being backed up. In an embodiment, the access control group is a local administrators group. In an embodiment, the user accounts each include one or more user identifiers.

[0028] User system **102** can connect to host system **104** via interface **108** and a user of user system **102** can log onto virtual machine **112.1**. When logged onto virtual machine **112.1**, the user of user system **102** can run a backup application user interface **202** on virtual machine **112.1**. Backup application user interface **202** is a user interface for backup application **204**. In an embodiment, backup application user interface **202** is a web-based user interface, which can be accessed using an Internet browser.

[0029] User of user system **102** can use backup application user interface **202** of virtual machine **112.1** to send a restoration request **206** to backup server **106**. In an embodiment, restoration request

206 includes a machine identifier of virtual machine **112.1** (on which backup application user interface **202** is running). The machine identifier can be a Domain Name System (DNS) name, an Internet Protocol (IP) address assigned to the virtual machine, a hypervisor-level VM identifier, or similar identifier. In another embodiment, the machine identifier can be automatically detected by matching a computer account (such as an Active Directory computer account) to a VM.

[0030] Backup application **204** is configured to receive restoration request **206** from virtual machine **112.1** over interface **116**. Using the machine identifier contained in the restoration request, backup application **204** identifies whether the machine identifier contained in restoration request **206** can be matched to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106**. In an embodiment, if the machine identifier contained in restoration request **206** can be matched to the machine identifier of the virtual machine present in virtual machine image level backup **114.1** stored on backup server **106**, the user who is currently logged on virtual machine **112.1** can access and restore objects (e.g., files, folders, directories, etc.) from virtual machine image level backup **114.1** of virtual machine **112.1** using user interface **202**, which may be a web interface.

[0031] Backup application **204** then accepts or denies restoration request **206** based at least in part on whether the machine identifier contained in restoration request **206** can be matched to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106**. In an embodiment, backup application **204** accepts the restoration request if machine identifier contained in restoration request **206** can be matched to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106** and denies the restoration request otherwise. A response (acceptance or denial) **208** to restoration request **206** is then sent to backup application user interface **202**. If response **208** is a denial, backup application user interface **202** displays a request denied message to the user. Otherwise, backup application user interface **202** provides an interface for accessing image level backup VM_1 Backup **114.1** of virtual machine **112.1**. The interface can include a view for selecting objects for restoration from image level backup VM_1 Backup **114.1**.

[0032] FIG. 3 is an example process **300** according to an embodiment. Example process **300** is provided for the purpose of illustration only and is not limiting of embodiments. Example process **300** can be performed by a backup server, such as backup server **106**, and more particularly a backup application, such as backup application **204**.

[0033] As shown in FIG. 3, example process **300** begins in step **302**, which includes saving, during backup or replication activity, to a database user accounts belonging to a certain access control group in the OS of Virtual Machines being backed up. In an embodiment, the access control group is local administrators group.

[0034] Subsequently, process **300** proceeds to step **304**, which includes receiving a restoration request from a virtual machine. In an embodiment, the restoration request includes machine identifier. The machine identifier can be DNS name, IP address assigned to virtual machine, hypervisor-level VM identifier, and similar. In another embodiment, VM identifier can be automatically detected by matching computer account (such as Active Directory computer account) to a VM.

[0035] Subsequently, process **300** proceeds to step **306**, which includes determining whether the machine identifier contained in restoration request **206** can be matched to the machine identifier of a virtual machine present in one of the virtual machine backups stored on backup server. If the answer is no, process **300** proceeds to step **308**, which includes denying the restoration request. Otherwise, process **300** proceeds to step **310**, which includes accepting the restoration, and then to step **312**, which includes providing the user access to content of an image level backup of the virtual machine.

[0036] FIG. 4 illustrates another example system **400** according to an embodiment. Example system **400** is provided for the purpose of illustration only and is not limiting of embodiments. As

shown in FIG. 4, example system 400 includes a user system 102, a host system 104, and a backup server 106. Host system 104 includes a hypervisor 110, which enables one or more virtual machines, such as virtual machine 112.1. Host system 104 is connected to user system 102 via an interface 108 and to backup server 106 via an interface 116.

[0037] As described above with respect to example system 200, backup server 106 stores image level backups of one or more virtual machines. For example, backup server 106 stores image level backup VM_1 Backup 114.1 of virtual machine 112.1. Backup server 106 also hosts a backup application 204, which can be used by users to backup virtual machines, such as virtual machine 112.1, to backup server 106 or to restore virtual machine images or portions thereof, from image level backup VM_1 Backup 114.1, from backup server 106.

[0038] In an embodiment, during backup or replication activity, backup application 204 saves to a database user accounts belonging to a certain access control group in the OS of Virtual Machines being backed up. In an embodiment, the access control group is a local administrators group

[0039] In an embodiment, a user of user system 102 can use backup application user interface 202 of virtual machine 112.1 to send a restoration request 402 to backup server 106. In an embodiment, restoration request 402 includes a machine identifier of virtual machine 112.1 (on which backup application user interface 202 is running), and a user identifier of the user logged onto virtual machine 112.1. The user identifier can be the login information of the user logged onto the virtual machine, a user token, or similar identifier.

[0040] In an embodiment, as described above with reference to FIG. 2, backup application 204 is configured to use the machine identifier of virtual machine 112.1 contained in restoration request 402 to match it to the machine identifier of virtual machine 112.1 present in virtual machine image level backup 114.1 stored on backup server 106. Backup application 204 then accepts or denies restoration request 402 based at least in part whether machine identifier contained in restoration request 206 can be matched to the machine identifier of virtual machine 112.1 present in virtual machine image level backup 114.1 stored on backup server 106. In an embodiment, backup application 204 denies restoration request 402 if the machine identifier contained in restoration request 206 cannot be matched to the machine identifier of virtual machine 112.1 present in virtual machine image level backup 114.1 stored on backup server 106

[0041] In an embodiment, if the machine identifier machine identifier contained in restoration request 206 can be matched to the machine identifier of virtual machine 112.1 present in virtual machine image level backup 114.1 stored on backup server 106, backup application 204 is then configured to determine whether or not the user identifier, contained in the restoration request, belongs to access control group 210 in the OS. In an embodiment, access control group 210 is a local administrators group. Backup application 204 then accepts or denies restoration request 402 based at least in part on whether or not the user identifier, contained in the request, belongs to access control group 210 of virtual machine 112.1. In an embodiment, backup application 204 denies restoration request 402 if the user identifier does not belong to access control group 210. A response 208 denying the restoration request is then sent to backup application user interface 202.

[0042] If response 208 includes a denial, response 208 can indicate to the user via backup application user interface 202 that the reason for denial is that user identifier does not belong to access control group 210. In an embodiment, access control group 210 is a local administrators group. Otherwise, if restoration request 402 is accepted, then backup application user interface 202 provides an interface for accessing the content of image level backup VM_1 Backup 114.1 of virtual machine 112.1. The interface can include a view for selecting objects for restoration from image level backup VM_1 Backup 114.1.

[0043] In an embodiment, the user identifier can be a user token. Backup application 204 uses the user token to determine whether or not the user identifier, contained in the restoration request, belongs to access control group 210. In an embodiment, the access control group is local administrators group. For example, in case of Microsoft Windows being an operating system of

virtual machine **111.2**, the user token contains information regarding SIDs of users group that the user belongs to.

[0044] FIG. **5** is another example process **500** according to an embodiment. Example process **500** is provided for the purpose of illustration only and is not limiting of embodiments. Example process **500** can be performed by a backup server, such as backup server **106**, and more particularly by a backup application, such as backup application **204**.

[0045] As shown in FIG. **5**, example process **500** begins in step **502**, which includes saving, during backup or replication activity, to a database user accounts belonging to a certain access control group in the OS of Virtual Machines being backed up. In an embodiment, the access control group is local administrators group.

[0046] Subsequently, example process **500** proceeds to in step **504**, which includes receiving a restoration request from a virtual machine. In an embodiment, the restoration request includes a machine identifier and a user identifier of a user logged onto the virtual machine. The machine identifier can be a DNS name, an IP address assigned to virtual machine, a hypervisor-level VM identifier, or similar identifier. In another embodiment, a VM identifier can be automatically detected by matching a computer account (such as Active Directory computer account) to a VM.

[0047] Subsequently, process **500** proceeds to step **506**, which includes determining whether the machine identifier contained in restoration request **206** can be matched to the machine identifier of a virtual machine present in one of the virtual machine image level backups stored on the backup server.

[0048] If the answer is no, process **500** proceeds to step **508**, which includes denying the restoration request. Otherwise, process **500** proceeds to step **510**.

[0049] In step **510**, process **500** includes determining whether or not the user identifier, contained in the restoration request, belongs to the a certain access control group in the OS of the virtual machine. In an embodiment, the access control group is a local administrators group. If the answer is no, process **500** proceeds to step **508**, which includes denying the restoration request. Otherwise, process **500** proceeds to step **512**, which includes accepting the restoration request, and then to step **514** which includes providing the user access to content of an image level backup of the virtual machine.

[0050] FIG. **6** illustrates another example system **600** according to an embodiment. In this embodiment, an authorization code can be delivered to the user by placing it directly into the virtual machine **112** file system to a folder accessible to specific user or computer administrator only, such as home folder. Example system **600** is provided for the purpose of illustration only and is not limiting of embodiments.

[0051] As shown in FIG. **6**, example system **600** includes a user system **102**, a host system **104**, and a backup server **106**. Host system **104** includes a hypervisor **110**, which enables one or more virtual machines, such as virtual machine **112.1**. Host system **104** is connected to user system **102** via an interface **108** and to backup server **106** via an interface **116**.

[0052] As described above with respect to example system **200**, backup server **106** stores image level backups of virtual machines. For example, backup server **106** stores image level backup VM_1 Backup **114.1** of virtual machine **112.1**. Backup server **106** also hosts a backup application **204**, which can be used by users to backup virtual machines, such as virtual machine **112.1**, to backup server **106** or to restore virtual machine images or portions thereof from image level backup VM_1Backup **114.1** stored on backup server **106**.

[0053] In an embodiment, a user of user system **102** can use backup application user interface **202** of virtual machine **112.1** to send a restoration request **206** to backup server **106**. In an embodiment, restoration request **206** includes a machine identifier **112.1** (on which backup application user interface **202** is running) and a user identifier of the user logged onto virtual machine **112.1**. The user identifier can be the login information of the user logged onto the virtual machine, user token, or similar.

[0054] In an embodiment, backup application **204** is configured to use the machine identifier of virtual machine **112.1** contained in restoration request **402** to match it to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106**. Backup application **204** then accepts or denies restoration request **206** based at least in part on whether the machine identifier contained in restoration request **206** can be matched to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106**. In an embodiment, backup application **204** denies restoration request **402** if the machine identifier contained in restoration request **206** cannot be matched to the machine identifier of virtual machine **112.1** present in virtual machine image level backup **114.1** stored on backup server **106**. A response **208** denying the restoration request is then sent to backup application user interface **202**.

[0055] If the machine identifier contained in restoration request **206** can be matched to the machine identifier of virtual machine **112.1** present virtual machine image level backup **114.1** stored on backup server **106**, backup application **204** is configured to write an authentication cookie **606** to an administrator-only accessible location **602** of virtual machine **112.1** and to prompt the user logged onto virtual machine **112.1** to provide the authentication cookie. In an embodiment, administrator-only accessible location **602** is a directory of virtual machine **112.1** that can only be accessed by users with administrative access privileges. A user authorized to access and restore objects (e.g., files, folders, directories, etc.) from image level backups of virtual machine **112.1** is part of this set of users and can therefore access and retrieve authentication cookie **606** from administration-only accessible location **602**. In an embodiment, backup application **204** uses an API **604** provided by hypervisor **110** to write authentication cookie **606** to administrator-only accessible location **602** of virtual machine **112.1**. For example, in the case of hypervisor **110** being a VMware® hypervisor, API **604** can be a VIX API, which provides a library for writing scripts and programs to manipulate virtual machines. In an embodiment, authentication cookie **606** can be written to administrator-only accessible location **602** of virtual machine **112.1** using the CreateTempFileInGuest operation of the VIX API, which copies a file or directory from backup server **106** to administrator-only accessible location **602** of virtual machine **112.1**.

[0056] Backup application **204** is then configured to wait for the user logged onto virtual machine **112.1** to provide the authentication cookie via backup application user interface **202**. If no authentication cookie is received by backup application **204** with a predetermined time interval, backup application **204** denies restoration request **206** and sends a denial response **208** to backup application user interface **202**. Response **208** can indicate to the user via backup application user interface **202** that the reason for the denial is the expiration of the time to enter the authentication.

[0057] Otherwise, if an authentication cookie **608** is received by backup application **204** within the predetermined time interval, then backup application **204** accepts restoration request **206**, if the received authentication cookie **608** matches the written authentication cookie **606** and denies restoration request **206**, if the received authentication cookie **608** does not match the written authentication cookie **606**. A response **208** accepting or denying restoration request **206** is then sent to backup application user interface **202**. If response **208** includes a denial, response **208** can indicate to the user via backup application user interface **202** that the reason for denial is the entry of an incorrect/invalid authentication cookie. Otherwise, if restoration request **206** is accepted, then backup application user interface **202** provides an interface for accessing content of image level backup VM_1 Backup **114.1** of virtual machine **112.1**. The interface can include a view for selecting objects for restoration from image level backup VM_1 Backup **114.1**.

[0058] FIG. 7 is another example process **700** according to an embodiment. Example process **700** is provided for the purpose of illustration only and is not limiting of embodiments. Example process **700** can be performed by a backup server, such as backup server **106**, and more particularly a backup application, such as backup application **204**. As shown in FIG. 7, example process **700** begins in step **702**, which includes receiving a restoration request from a virtual machine. In an

embodiment, the restoration request includes a machine identifier and a user identifier of a user logged onto the virtual machine. Subsequently, in step **704**, process **700** includes determining whether the machine identifier contained in the restoration request can be matched to the machine identifier of a virtual machine present in one of the virtual machine image level backups stored on the backup server. If the answer is no, process **700** proceeds to step **706**, which includes denying the restoration request. Otherwise, process **700** proceeds to step **708**.

[0059] In step **708**, process **700** includes writing an authentication cookie to an administrator-only accessible location of the virtual machine. In an embodiment, the administrator-only accessible location is a directory of the virtual machine that can only be accessed by users with administrative access privileges. In an embodiment, the authentication cookie is written to the virtual machine using a VIX API provided by a VMware® hypervisor.

[0060] Subsequently, in step **710**, process **700** includes prompting the user logged onto the virtual machine to provide the authentication cookie. A user authorized to access and restore objects (e.g., files, folders, directories, etc.) from image level backups of the virtual machine is part of the set of users that can access the administrator-only accessible location of the virtual machine, and can therefore access and retrieve the authentication cookie from the administrator-only accessible location.

[0061] Then, process **700** proceeds to step **712**, in which the backup applications waits to receive an authentication cookie from the virtual machine. If no authentication cookie is received from the virtual machine within a predetermined time interval, process **700** proceeds to step **706**, which includes denying the restoration request. Otherwise, process **700** proceeds to step **714**.

[0062] In step **714**, process **700** includes determining whether the received authentication cookie matches the written authorization code. If the is answer is no, process **700** proceeds to step **706**. Otherwise, process **700** proceeds to step **716**, which includes accepting the restoration request, and then to step **718**, which includes providing the user access to content of image level backup of the virtual machine.

[0063] FIG. **8** illustrates another example system **800** according to an embodiment. Example system **800** is provided for the purpose of illustration only and is not limiting of embodiments. As shown in FIG. **8**, example system **800** includes a user system **102**, a host system **104**, and a backup server **106**. Host system **104** includes a hypervisor **110**, which enables one or more virtual machines, such as virtual machine **112.1**, **112.2**, . . . , **112.N**. Host system **104** is connected to user system **102** via an interface **108** and to backup server **106** via an interface **116**.

[0064] As described above with respect to example system **200**, backup server **106** stores image level backups of one or more virtual machines. For example, backup server **106** can store image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N** respectively. Backup server **106** also hosts a backup application **204**, which can be used by users to backup virtual machines, such as virtual machine **112.1**, **112.2**, . . . , **112.N** to backup server **106** or to restore virtual machine images or portions thereof, from image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N** respectively, from backup server **106**.

[0065] In an embodiment, during backup or replication activity, backup application **204** saves to a database user accounts belonging to a certain access control group in the OS of Virtual Machines being backed up. In an embodiment, the access control group is a local administrators group.

[0066] In an embodiment, a user of user system **102** can use backup application user interface **202.1**, **202.2**, . . . , **202.N** of virtual machines **112.1**, **112.2**, . . . , **112.N** to send a restoration request **802** to backup server **106**. In an embodiment, restoration request **802** includes a machine identifier of virtual machine **112.1** (on which backup application user interface **202** is running), and a user identifier of the user logged onto virtual machine **112.1**. The user identifier can be the login information of the user logged onto the virtual machine, a user token, or similar identifier.

[0067] In an embodiment, backup application **204** is configured to use the user identifier contained in the restoration request to match it to one of the user identifiers contained in the image level

backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N**. Backup application **204** then accepts or denies restoration request **802** based at least in part on whether or not the user identifier contained in the restoration request can be matched to one or more of the user identifiers contained in the image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N**. In an embodiment, backup application **204** denies restoration request **402** if the user identifier contained in the restoration request cannot be matched to any of the user identifiers contained in the image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N**. A response **208** denying the restoration request is then sent to backup application user interface **202**.

[0068] If response **208** includes a denial, response **208** can indicate to the user via backup application user interface **202** that the reason for denial is the user identifier contained in the restoration request cannot be matched to any of the user identifiers contained in the virtual machine image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N** respectively. Otherwise, if restoration request **402** is accepted, then backup application user interface **202** provides an interface for accessing the content of virtual machine image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machine **112.1**, **112.2**, . . . , **112.N**. The interface can include a view for selecting objects for restoration from image level backups **114.1**, **114.2**, . . . , **114.N** of virtual machines **112.1**, **112.2**, . . . , **112.N**.

[0069] In an embodiment, backup application **204** can use the machine identifier contained in the restoration request to limit visible scope of content of image level backups. For example, using machine identifier contained in the restoration request, the backup application **204** can limit visible scope of image level backups to content of image level backup corresponding to the VM that the restoration request came from.

[0070] FIG. **9** is an example process **900** according to an embodiment. Example process **900** is provided for the purpose of illustration only and is not limiting of embodiments. Example process **900** can be performed by a backup server, such as backup server **106**, and more particularly a backup application, such as backup application **204**.

[0071] As shown in FIG. **9**, example process **900** begins in step **902**, which includes saving, during backup or replication activity, to a database user accounts belonging to a certain access control group in the OS of Virtual Machines being backed up. In an embodiment, the access control group is local administrators group.

[0072] Subsequently, process **900** proceeds to step **904**, which includes receiving a restoration request from a virtual machine. In an embodiment, the restoration request includes a machine identifier. The machine identifier can be a DNS name, an IP address assigned to virtual machine, a hypervisor-level VM identifier, or similar identifier. In another embodiment, a VM identifier can be automatically detected by matching a computer account (such as Active Directory computer account) to a VM. In an embodiment, the restoration request includes a user identifier of the user logged onto virtual machine. The user identifier can be the login information of the user logged onto the virtual machine, a user token, or similar identifier.

[0073] Subsequently, process **900** proceeds to step **906**, which includes determining whether or not the user identifier contained in the restoration request can be matched to one or more of the user identifiers contained in the image level backups.

[0074] If the answer is no, process **900** proceeds to step **908**, which includes denying the restoration request. Otherwise, process **900** proceeds to step **910**, which includes accepting the restoration request, and then to step **912** which includes providing the user access to content of an image level backup of the virtual machine.

[0075] In step **914**, process **900** includes using the machine identifier contained in the restoration request to limit visible scope of content of image level backups.

[0076] Various aspects of the embodiments described herein can be implemented by software, firmware, hardware, or a combination thereof. FIG. **10** illustrates an example computer system

1000 in which embodiments, or portions thereof, can be practiced or implemented as computer-readable code. For example, processes **300** of FIG. 3, **500** of FIG. 5, **700** of FIG. 7, and **900** of FIG. 9 can be implemented in system **1000**. Various embodiments are described in terms of this example computer system **1000**. After reading this description, it will become apparent to a person skilled in the relevant art how to implement embodiments using other computer systems and/or computer architectures.

[0077] Computer system **1000** includes one or more processors, such as processor **1010**. Processor **1010** can be a special-purpose or a general-purpose processor. Processor **1010** is connected to a communication infrastructure **1020** (for example, a bus or network). Computer system **1000** also includes a main memory **1030**, preferably random access memory (RAM), and may also include a secondary memory **1040**. Secondary memory **1040** may include, for example, a hard disk drive **1050**, a removable storage drive **1060**, and/or a memory stick. Removable storage drive **1060** may comprise a floppy disk drive, a magnetic tape drive, an optical disk drive, a flash memory, or the like. Removable storage drive **1060** reads from and/or writes to a removable storage unit **1070** in a well-known manner. Removable storage unit **1070** may comprise a floppy disk, magnetic tape, optical disk, etc. As will be appreciated by persons skilled in the relevant art(s), removable storage unit **1070** includes a computer-usable storage medium having stored therein computer software and/or data.

[0078] In alternative implementations, secondary memory **1040** may include other similar means for allowing computer programs or other instructions to be loaded into computer system **1000**. Such means may include, for example, a removable storage unit **1070** and an interface **1020**. Examples of such means may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM, or PROM) and associated socket, and other removable storage units **1070** and interfaces **1020** which allow software and data to be transferred from the removable storage unit **1070** to computer system **1000**.

[0079] Computer system **1000** may also include a communication and network interface **1080**. Communication interface **1080** allows software and data to be transferred between computer system **1000** and external devices. Communication interface **1080** may include a modem, a communication port, a PCMCIA slot and card, or the like. Software and data transferred via communication interface **1080** are in the form of signals which may be electronic, electromagnetic, optical, or other signals capable of being received by communication interface **1080**. These signals are provided to communication interface **1080** via a communication path **1085**. Communication path **1085** carries signals and may be implemented using wire or cable, fiber optics, a phone line, a cellular phone link, an RF link or other communication channels.

[0080] The network interface **1080** allows the computer system **1000** to communicate over communication networks or mediums such as LANs, WANs the Internet, etc. The network interface **1080** may interface with remote sites or networks via wired or wireless connections. The computer system **1000** may also include input/output/display devices **1090**, such as keyboards, monitors, pointing devices, etc.

[0081] In this document, the terms “computer readable medium” and “computer usable medium” are used to generally refer to media such as removable storage unit **1070**, removable storage drive **1060**, and a hard disk installed in hard disk drive **1050**. Computer program medium and computer usable medium can also refer to memories, such as main memory **1030** and secondary memory **1040**, which can be memory semiconductors (e.g. DRAMs, etc.). These computer program products are means for providing software to computer system **1000**.

[0082] Computer programs (also called computer control logic) are stored in main memory **1030** and/or secondary memory **1040**. Computer programs may also be received via communication interface **1080**. Such computer programs, when executed, enable computer system **1000** to implement the embodiments discussed herein. In particular, the computer programs, when executed, enable processor **1010** to implement the processes discussed above in FIGS. 3, 5, and 9.

Accordingly, such computer programs represent controllers of the computer system **1000**. Where an embodiment is implemented using software, the software may be stored in a computer program product and loaded into computer system **1000** using removable storage drive **1060**, interface **1010**, hard disk drive **1050** or communication interface **1080**.

[0083] Embodiments have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

CONCLUSION

[0084] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections (if any), is intended to be used to interpret the claims. The Summary and Abstract sections (if any) may set forth one or more but not all exemplary embodiments of the invention as contemplated by the inventor(s), and thus, are not intended to limit the invention or the appended claims in any way.

[0085] While the invention has been described herein with reference to exemplary embodiments for exemplary fields and applications, it should be understood that the invention is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of the invention. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

[0086] Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments may perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

[0087] References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein.

[0088] The breadth and scope of the invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A system communicatively coupled with a user system and a backup server, the system comprising: a processor; and a hypervisor configured to enable a virtual machine implementing a backup application user interface, wherein the backup application user interface is configured to provide access to the backup server, and wherein the hypervisor is configured to: log the user system onto the virtual machine using a user identifier; receive, at the backup application user interface from the user system, a restoration request, wherein the restoration request includes the user identifier and a machine identifier of the virtual machine, wherein the restoration request is

configured to request restoring an image level backup of the virtual machine stored on the backup server; transmit, by the backup application user interface to the backup server, the restoration request to the backup server; transmit, by the backup application user interface to the user system, a prompt to provide additional authentication prior to providing an interface to the user system, wherein the interface is configured to provide access the image level backup of the virtual machine; and provide, via the backup application user interface, the interface to the user system, wherein the interface is provided responsive to receiving an acceptance of the restoration request from the backup server.

2. The system of claim 1, wherein the interface comprises a graphical view for selecting one or more objects for restoration from the image level backup.

3. The system of claim 1, wherein the hypervisor is further configured to: receive, at the backup application user interface from a second user system, a second restoration request, wherein the second restoration request is configured to request restoring the image level backup of the virtual machine stored on the backup server; and receive, from the backup server, a rejection of the second restoration request from the backup server, wherein the rejection of the second restoration request comprises a message indicating the additional authentication is incorrect.

4. The system of claim 1, wherein the machine identifier comprises one of a domain name system name, an Internet Protocol address assigned to the virtual machine, or a hypervisor-level virtual machine identifier.

5. The system of claim 1, wherein the hypervisor is configured to store an access control group, wherein the interface to access is further provided based on determining that the user identifier belongs to the access control group.

6. The system of claim 5, wherein the access control group comprises a local administrator group.

7. The system of claim 5, wherein the user identifier is a user token containing information regarding at least one identifier of a user group to which the user system is assigned.

8. The system of claim 1, wherein the system comprises a second virtual machine, wherein the second virtual machine comprises a second backup application user interface.

9. The system of claim 1, wherein the additional authentication comprises an authentication cookie, and wherein the virtual machine further comprises an administrator-only accessible storage location, and wherein the hypervisor comprises an application programming interface configured to communicate with the backup server.

10. The system of claim 9, wherein the authentication cookie is generated responsive to the restoration request being accepted by the backup server based on the machine identifier of the virtual machine, and wherein the hypervisor is further configured to: write, to the administrator-only accessible storage location, the authentication cookie.

11. The system of claim 10, wherein the interface is further provided responsive to the hypervisor transmitting, via the backup application user interface, the authentication cookie to the backup server within a predetermined time interval.

12. A computer-implemented method to provide a virtual machine implementing a backup application user interface access to a backup server, comprising: logging a user system onto the virtual machine using a user identifier; receiving, at the backup application user interface from the user system, a restoration request, wherein the restoration request includes the user identifier and a machine identifier of the virtual machine, wherein the restoration request is configured to request restoring an image level backup of the virtual machine stored on the backup server; transmitting, by the backup application user interface to the backup server, the restoration request to the backup server; transmitting, by the backup application user interface to the user system, a prompt to provide additional authentication prior to providing an interface to the user system, wherein the interface is configured to provide access the image level backup of the virtual machine; and providing, via the backup application user interface, the interface to the user system, wherein the interface is provided responsive to receiving an acceptance of the restoration request from the

backup server.

13. The computer-implemented method of claim 12, wherein the interface comprises a graphical view for selecting one or more objects for restoration from the image level backup.

14. The computer-implemented method of claim 12, the method comprising: receiving, at the backup application user interface from a second user system, a second restoration request, wherein the second restoration request is configured to request restoring the image level backup of the virtual machine stored on the backup server; and receiving, from the backup server, a rejection of the second restoration request from the backup server, wherein the rejection of the second restoration request comprises a message indicating the additional authentication is incorrect.

15. The computer-implemented method of claim 12, wherein the machine identifier comprises one of a domain name system name, an Internet Protocol address assigned to the virtual machine, or a hypervisor-level virtual machine identifier.

16. The computer-implemented method of claim 12, the operations further comprising storing an access control group, wherein the interface to access is further provided based on determining that the user identifier belongs to the access control group.

17. The method of claim 16, wherein the access control group comprises a local administrator group.

18. The computer-implemented method of claim 16, wherein the user identifier is a user token containing information regarding at least one identifier of a user group to which the user is assigned.

19. The computer-implemented method of claim 12, wherein the additional authentication comprises an authentication cookie, and wherein the virtual machine further comprises an administrator-only accessible storage location, and wherein the hypervisor comprises an application programming interface configured to communicate with the backup server.

20. A computer program product comprising a non-transitory computer readable medium including computer control logic stored therein, wherein the computer control logic, when executed by one or more processors, enables operations for providing a virtual machine implementing a backup application user interface access to a backup server, the operations comprising: logging a user system onto the virtual machine using a user identifier; receive, at the backup application user interface from the user system, a restoration request, wherein the restoration request includes the user identifier and a machine identifier of the virtual machine, wherein the restoration request is configured to request restoring an image level backup of the virtual machine stored on the backup server; transmit, by the backup application user interface to the backup server, the restoration request to the backup server; transmit, by the backup application user interface to the user system, a prompt to provide additional authentication prior to providing an interface to the user system, wherein the interface is configured to provide access the image level backup of the virtual machine; and provide, via the backup application user interface, the interface to the user system, wherein the interface is provided responsive to receiving an acceptance of the restoration request from the backup server.
